Mixture-of-Experts for Object Detection (MoE in YOLO)

Literature Survey

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The Mixture-of-Experts (MoE) paradigm introduces conditional computation by routing each input to a small subset of specialized sub-networks rather than processing it through the entire model. The modern formulation originates from the sparsely-gated MoE layer of Shazeer et al. [1], which demonstrated that a learned gating network selecting the top-k experts enables large increases in model capacity at near-constant computational cost. Fedus et al. [2] subsequently simplified and stabilized this mechanism in the Switch Transformer, establishing top-1 routing as a scalable and practical design.

While MoE was first established in natural language processing, its extension to computer vision was shown by Riquelme et al. in V-MoE [3]. V-MoE replaces a subset of the dense feed-forward layers of the Vision Transformer with sparsely activated expert layers, matching the accuracy of state-of-the-art dense networks while requiring as little as half of the inference-time compute. This work provides the principal evidence that sparse expert routing transfers effectively to visual recognition tasks.

The application of MoE specifically to object detection remains comparatively underexplored. Two representative peer-reviewed directions are relevant. DAMEX [4] routes inputs according to dataset identity to enable unified detection across a mixture of datasets, illustrating expert specialization driven by data provenance. More recently, Dynamic-DINO [5] incorporates fine-grained MoE tuning into a real-time open-vocabulary detector; in its shallow layers experts cooperate with diverse peers to broaden the search space, and at inference only the input-relevant experts are activated to form a compact sub-network. To mitigate degradation at the onset of fine-tuning, it further employs a pre-trained weight allocation strategy for the experts together with a dedicated router initialization.

In summary, prior work establishes that sparse MoE layers scale effectively in both language and vision, yet their integration into efficient single-stage detectors such as YOLO has received limited attention. The present project addresses this gap by inserting an MoE module with top-k gating into a YOLO backbone and neck, and evaluating whether conditional expert routing improves detection performance relative to the baseline detector under a constrained computational budget.

References

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